IOT ASSIGNMENT

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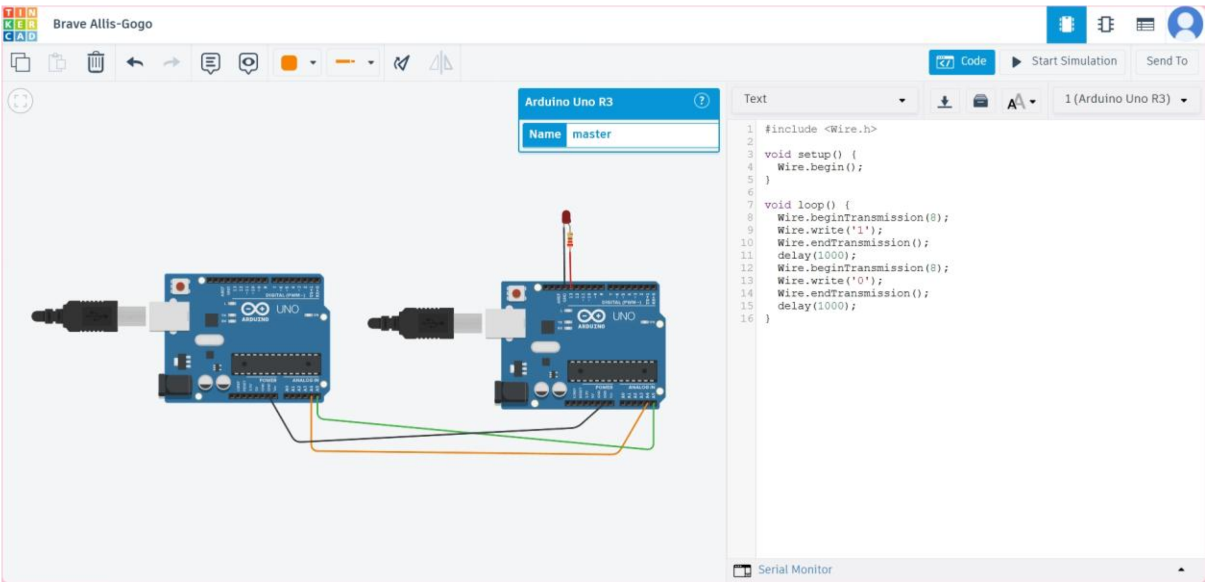
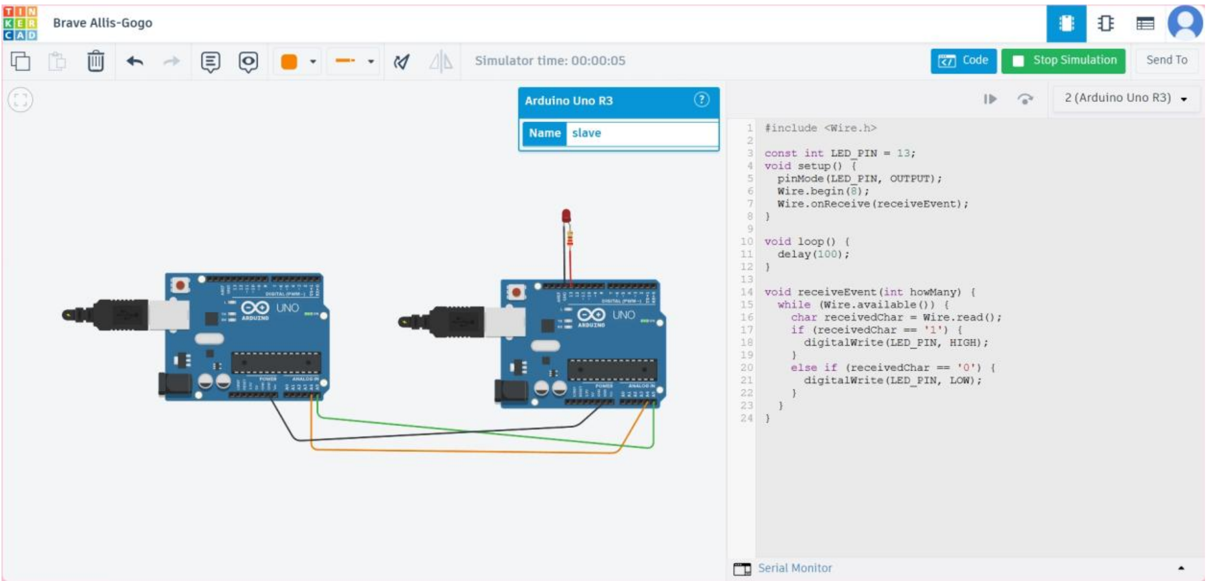
Q1

```
const int sleep_seconds = 10;

void setup() {
  Serial.begin(115200);
  delay(1000);
  Serial.println("ESP32 woken up");
  esp_sleep_enable_timer_wakeup(sleep_seconds * 1000000);

  Serial.print("Going into deep sleep mode for ");
  Serial.print(sleep_seconds);
  Serial.println(" seconds...");
  Serial.flush();
  esp_deep_sleep_start();
}

void loop() {
}
```

Q2.

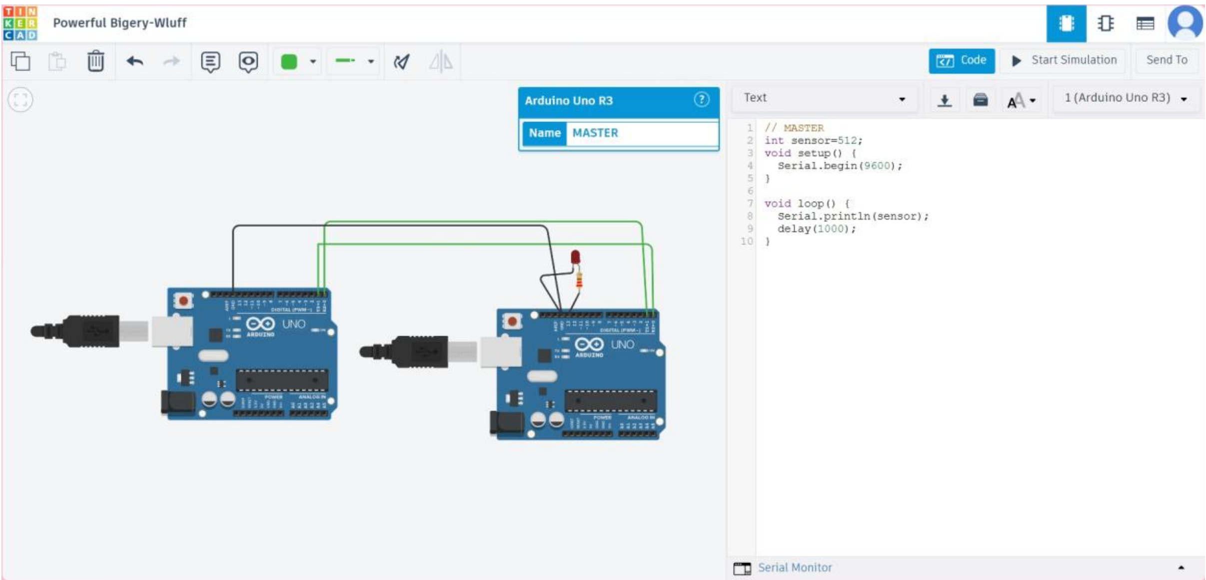




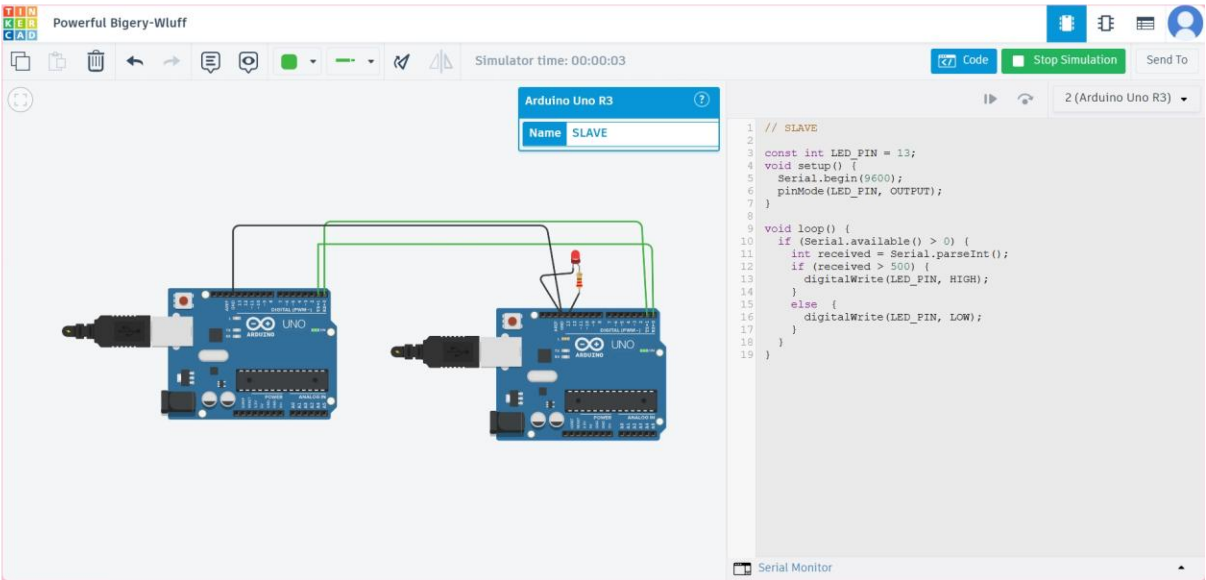
Q3.

```
#include <SPI.h>
const int CS_PIN = 5;

void setup() {
  pinMode(CS_PIN, OUTPUT);
  digitalWrite(CS_PIN, HIGH);
  SPI.begin();
  SPI.beginTransaction(SPISettings(1000000, MSBFIRST, SPI_MODE0));
}

void loop() {
  digitalWrite(CS_PIN, LOW);
  SPI.transfer(0x55);
  digitalWrite(CS_PIN, HIGH);
  delay(1000);
}
```

Q4.


Powerful Bigery-Wluff
Code
Start Simulation
Send To
Arduino Uno R3
Name MASTER
Text
1 (Arduino Uno R3)
// MASTER
int sensor=512;
void setup() {
  Serial.begin(9600);
}

void loop() {
  Serial.println(sensor);
  delay(1000);
}
Serial Monitor


Powerful Bigery-Wluff
Simulator time: 00:00:03
Code
Stop Simulation
Send To
Arduino Uno R3
Name SLAVE
2 (Arduino Uno R3)
// SLAVE

const int LED_PIN = 13;
void setup() {
  Serial.begin(9600);
  pinMode(LED_PIN, OUTPUT);
}

void loop() {
  if (Serial.available() > 0) {
    int received = Serial.parseInt();
    if (received > 500) {
      digitalWrite(LED_PIN, HIGH);
    }
    else {
      digitalWrite(LED_PIN, LOW);
    }
  }
}
Serial Monitor